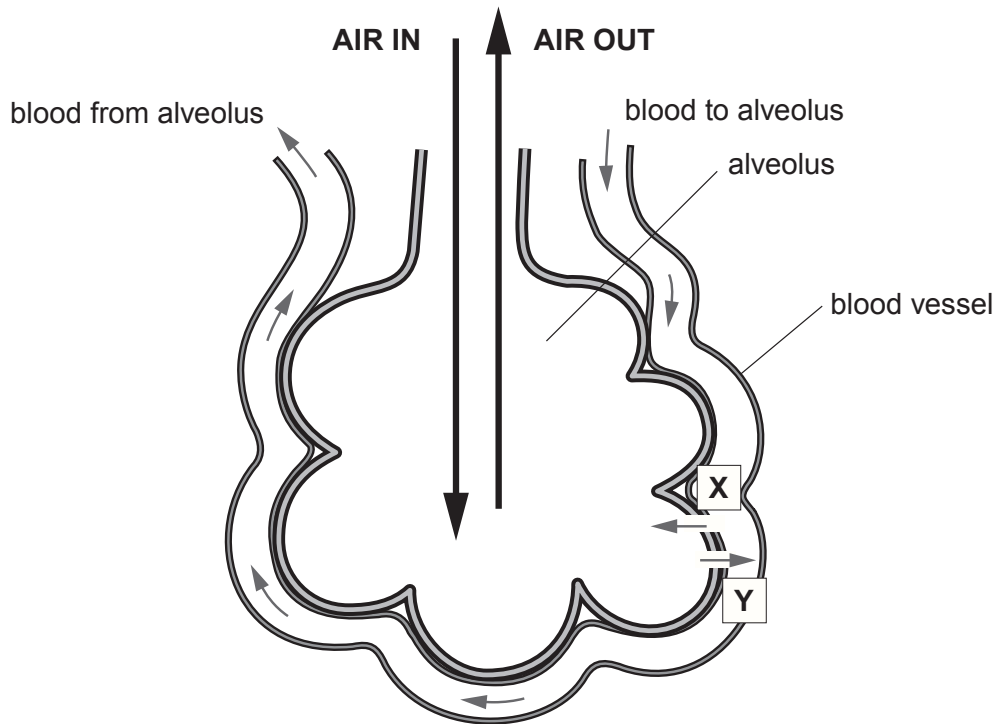


6. The human body needs a respiratory system for gas exchange.

(a) The diagram below shows an alveolus.



- (i) Gases **X** and **Y** diffuse into and out of the alveolus. Name the gases **X** and **Y** in the diagram.

[1]

X

Y

- (ii) Name the type of blood vessel that surrounds the alveolus.

[1]

.....



- (b) Jade is an Olympic athlete. She trains on a regular basis. Training improves her respiratory system.

The table shows information about Jade's breathing at rest and at exercise during her latest test.

	Breathing rate (breaths per minute)	Depth of breathing (cm ³ per breath)	Total volume of air breathed in (cm ³ per minute)
at rest	12	500	
during exercise	40		40 000

- (i) Complete the table. [2]

- (ii) Describe the changes that exercise has on Jade's breathing. [2]

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- (iii) Jade's coach wants her lungs to be able to absorb at least 2000 cm³ of oxygen per minute as this will help with aerobic respiration in her muscles.

Inhaled air contains 21% oxygen and exhaled air contains 16% oxygen.

Jade thinks she has achieved this in her latest test.

- Use the information given to determine whether you agree with Jade. [3]

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(c) The oxygen absorbed is used in the cells to release energy by aerobic respiration.

- (i) During intense exercise some of Jade's muscles use the process of **anaerobic** respiration. State what is meant by the term 'anaerobic respiration'. [1]

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- (ii) Give **one advantage** and **one disadvantage** of anaerobic respiration occurring in Jade's muscle cells. [2]

Advantage:

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Disadvantage:

.....

